

E224

Global and Population Health

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A public health nurse in a county with a new community college reviews diabetes data. On January 1, 2,000 of the county's 50,000 residents had diabetes. During the year, 240 new cases were diagnosed. The county fair drew record crowds that summer. What was the one-year incidence?

- A. 40 per 1,000
- B. 44.8 per 1,000
- C. 5.0 per 1,000 at risk
- D. 4.8 per 1,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

2. [TRANSFER — reversed stem / different variable]

In a town of 10,000 people, 300 currently have asthma. What is the point prevalence?

- A. It cannot be found without the number of new cases
- B. 33.3 per 1,000
- C. About 31 per 1,000
- D. 30 per 1,000 (3%)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

3.

A school nurse in a district that recently won a robotics championship investigates a norovirus outbreak. Sick students all used a salad bar whose sneeze guard was broken. The principal is new this year. In the epidemiologic triangle, the broken sneeze guard is which element?

- A. Environment
- B. Host
- C. Agent
- D. Vector, since anything that spreads disease is a vector

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

4.

A nurse at a clinic that recently extended its weekend hours finds that a patient with poorly controlled diabetes lives in an area with no store selling fresh food within 10 miles. The patient is a retired firefighter. Which SDOH domain is most directly involved?

- A. Economic stability
- B. Neighborhood and built environment
- C. Health care access and quality
- D. Individual behavior, not an SDOH

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

At a mass-casualty bus crash near a county fairground, a nurse uses START triage in light rain. An adult cannot walk, breathes 24 times a minute, has a radial pulse with capillary refill of 1.5 seconds, and follows simple commands. He is wearing a sports jersey. Which tag applies?

- A. Yellow (delayed)
- B. Black (expectant)
- C. Red (immediate)
- D. Green (minor)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Under START, an adult who cannot walk is breathing 34 times a minute. Which tag applies?

- A. Green (minor)
- B. Red (immediate)
- C. Black (expectant)
- D. Yellow (delayed)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

At a wedding reception at a lakeside venue with a live band, 150 guests ate the chicken salad and 60 became ill. Of 100 guests who did not eat it, 5 became ill. The caterer has 20 years of experience. What is the attack rate among guests who ate the chicken salad?

- A. 26%
- B. 40%
- C. 60%
- D. 5%

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

Using the same outbreak (60 of 150 who ate were ill; 5 of 100 who did not eat were ill), what is the relative risk of illness for those who ate the chicken salad?

- A. 35
- B. 12
- C. 0.125
- D. 8.0

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

A hospital in a region with several new retirement communities joins a value-based contract that shares savings if its heart-failure patients have fewer readmissions and ED visits while meeting quality scores. The contract was signed at an annual conference. Which nursing action best supports the contract's goals?

- A. Discharge patients faster regardless of readiness
- B. Increase the number of billable procedures
- C. Follow-up calls, medication reconciliation, and home monitoring after discharge
- D. Focus only on cutting costs, since cost is all that counts

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A community health nurse in a city with a renovated riverfront visits a family with a 2-year-old. They live in a house built in 1955 with peeling paint on the windowsills. The father is a professional musician. What is the priority environmental concern and action?

- A.** Radon; recommend a basement radon test
- B.** None, since the lead-paint ban made all homes safe
- C.** Asbestos; tell the family to remove the windowsills themselves
- D.** Possible lead exposure; arrange blood lead testing and teach lead-safe cleaning

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A county with a new regional airport reports a measles case. Measles has an R0 of about 15. The county health officer is a former pediatrician. About what share of the population must be immune to reach the herd immunity threshold?

- A.** About 93%
- B.** About 7%
- C.** About 85%
- D.** 100%, since herd immunity requires everyone

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A disease has a herd immunity threshold of 75%. What is its R0?

- A.** 0.25
- B.** 75
- C.** 4
- D.** 1.33

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A public health nurse in a county ~~with a new community college~~ reviews diabetes data. On January 1, 2,000 of the county's 50,000 residents had diabetes. During the year, 240 new cases were diagnosed. ~~The county fair drew record crowds that summer.~~ What was the one-year incidence?

Target word: incidence **Decides:** $240 \div 48,000$ at risk.

A	Confusion set	That is the prevalence on January 1.
B	Adjacent category	That is end-of-year prevalence, $(2,000 + 240) \div 50,000$.
C	Correct	$240 \div (50,000 - 2,000) = 240 \div 48,000 = 5.0$ per 1,000.
D	Dropped element	Drops the existing cases from the denominator and uses 50,000.

2. Answer D Concept 1 • transfer

In a town of 10,000 people, 300 currently have asthma. What is the point prevalence?

Target word: prevalence **Decides:** $300 \div 10,000$.

A	Adjacent category	Prevalence counts existing cases; new cases are for incidence.
B	Inverted operation	Divides 10,000 by 300 instead of 300 by 10,000.
C	Confusion set	Uses the at-risk population (9,700), which belongs to incidence.
D	Correct	$300 \div 10,000 = 3\%$.

3. Answer A Concept 2

A school nurse in a district ~~that recently won a robotics championship~~ investigates a norovirus outbreak. Sick students all used a salad bar whose sneeze guard was broken. ~~The principal is new this year.~~ In the epidemiologic triangle, the broken sneeze guard is which element?

Target word: triangle element **Decides:** A condition that allows exposure.

A	Correct	Conditions that allow the agent to reach the host are environmental.
B	Adjacent category	The hosts are the students.
C	Confusion set	The agent is the norovirus itself.
D	Absolute language	A vector is a living carrier such as a mosquito.

4. Answer B Concept 3

A nurse at a clinic ~~that recently extended its weekend hours~~ finds that a patient with poorly controlled diabetes lives in an area with no store selling fresh food within 10 miles. ~~The patient is a retired firefighter.~~ Which SDOH domain is most directly involved?

Target word: SDOH domain **Decides:** Food access where the patient lives.

A	Adjacent category	Income is not the barrier described.
B	Correct	Access to healthy food where a person lives is part of the built environment.
C	Confusion set	The clinic is available; the barrier is food access.
D	Inverted operation	Reverses the point; the barrier is in the environment, not the person.

5. Answer A Concept 4

At a mass-casualty bus crash near a county fairground, a nurse uses START triage in light rain. An adult cannot walk, breathes 24 times a minute, has a radial pulse with capillary refill of 1.5 seconds, and follows simple commands. ~~He is wearing a sports jersey.~~ Which tag applies?

Target word: START tag **Decides:** Can't walk, but RPM all within limits.

A	Correct	Not walking, but breathing under 30, perfusion normal, and following commands.
B	Inverted operation	Black is for no breathing after the airway is opened.
C	Confusion set	Red requires RR over 30, poor perfusion, or failure to follow commands.
D	Dropped element	Drops the fact that he cannot walk.

6. Answer B Concept 4 • transfer

Under START, an adult who cannot walk is breathing 34 times a minute. Which tag applies?

Target word: START tag **Decides:** RR over 30.

A	Dropped element	Drops that the patient cannot walk.
B	Correct	A respiratory rate over 30 is an immediate tag.
C	Inverted operation	The patient is breathing; black is for no breathing after repositioning.
D	Wrong stem number	Misreads the cutoff; 34 is over 30.

7. Answer B Concept 5

At a wedding reception ~~at a lakeside venue with a live band~~, 150 guests ate the chicken salad and 60 became ill. Of 100 guests who did not eat it, 5 became ill. ~~The caterer has 20 years of experience.~~ What is the attack rate among guests who ate the chicken salad?

Target word: attack rate in exposed **Decides:** $60 \div 150$.

A	Dropped element	Uses all 250 guests and all 65 ill, dropping the exposure split.
B	Correct	$60 \div 150 = 40\%$.
C	Wrong stem number	Reports the count of ill guests as a percent.
D	Confusion set	That is the attack rate among those who did not eat it.

8. Answer D Concept 5 • transfer

Using the same outbreak (60 of 150 who ate were ill; 5 of 100 who did not eat were ill), what is the relative risk of illness for those who ate the chicken salad?

Target word: relative risk **Decides:** $40\% \div 5\%$.

A	Adjacent category	That is the difference in percentage points, not a ratio.
B	Wrong stem number	Divides the raw counts ($60 \div 5$) instead of the rates.
C	Inverted operation	Divides the unexposed rate by the exposed rate.
D	Correct	$40\% \div 5\% = 8.0$.

9. Answer C Concept 6

A hospital in a region ~~with several new retirement communities~~ joins a value-based contract that shares savings if its heart-failure patients have fewer readmissions and ED visits while meeting quality scores. ~~The contract was signed at an annual conference.~~ Which nursing action best supports the contract's goals?

Target word: best action **Decides:** Rewards outcomes and lower total cost.

A	Dropped element	Drops the quality requirement and risks readmission.
B	Inverted operation	That is a fee-for-service volume incentive.
C	Correct	Strong transitions prevent readmissions and ED visits.
D	Absolute language	Value-based contracts also require quality.

10. Answer D Concept 7

A community health nurse in a city ~~with a renovated riverfront~~ visits a family with a 2-year-old. They live in a house built in 1955 with peeling paint on the windowsills. ~~The father is a professional musician.~~ What is the priority environmental concern and action?

Target word: priority concern and action **Decides:** Pre-1978 home, peeling paint, young child.

A	Adjacent category	Radon is a real concern but not the one the peeling paint points to.
B	Absolute language	The 1978 ban did not remove lead paint already in older homes.
C	Confusion set	Asbestos is a different hazard, and do-it-yourself removal spreads dust.
D	Correct	Deteriorating paint in older homes is a key lead source for young children.

11. Answer A Concept 8

A county ~~with a new regional airport~~ reports a measles case. Measles has an R0 of about 15. ~~The county health officer is a former pediatrician.~~ About what share of the population must be immune to reach the herd immunity threshold?

Target word: threshold **Decides:** $1 - 1/15$.

A	Correct	$1 - 1/15 = 0.933$.
B	Inverted operation	That is $1 \div R0$ without subtracting from 1.
C	Wrong stem number	Treats R0 as a percentage to subtract from 100.
D	Absolute language	The threshold is below 100%.

12. Answer C Concept 8 • transfer

A disease has a herd immunity threshold of 75%. What is its R0?

Target word: R0 **Decides:** $1 \div (1 - 0.75)$.

A	Dropped element	Drops the final inversion step.
B	Wrong stem number	Uses the threshold percentage as the R0.
C	Correct	$1 - 1/4 = 0.75$.
D	Inverted operation	Divides 1 by 0.75 instead of by 0.25.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Incidence vs. prevalence	Incidence = new cases ÷ population at risk over a period; prevalence = all existing cases ÷ total population at a point or period.	Using the total population, including existing cases, as the incidence denominator.	“New cases” versus “currently have.”
2	Epidemiologic triangle	Disease results from the interaction of agent (the cause), host (the person), and environment (conditions that allow exposure).	Calling any contaminated object the agent.	A pathogen, susceptible people, and a setting factor.
3	Social determinants of health	SDOH domains are economic stability, education, health care access and quality, neighborhood and built environment, and social and community context.	Filing every barrier under health care access or individual choice.	A barrier rooted in where the person lives, works, or learns.
4	START mass-casualty triage	Walking = green; not breathing after airway opened = black; RR over 30, no radial pulse or refill over 2 seconds, or unable to follow commands = red; otherwise yellow.	Tagging every non-walking patient red, or missing the respiratory cutoff.	Walking status, respiratory rate, perfusion, and mental status.
5	Attack rate and relative risk	Attack rate = ill ÷ exposed in that group; relative risk = attack rate among exposed ÷ attack rate among unexposed.	Using everyone at the event as the denominator.	An outbreak table of ate/did not eat and ill/well.
6	Value-based care	Value-based payment rewards quality outcomes and lower total cost per population, so prevention and care transitions matter.	Treating it as a volume or speed incentive.	A contract tied to readmissions, quality scores, or shared savings.
7	Environmental health: lead	Children under 6 in homes built before 1978 with deteriorating paint are at high risk of lead exposure; screen blood lead and teach lead-safe practices.	Assuming the lead-paint ban made all older homes safe.	An older home, peeling paint, and a young child.
8	Herd immunity threshold	Herd immunity threshold = $1 - 1 \div R_0$; more contagious diseases need higher immunity.	Using $1 \div R_0$ or assuming everyone must be immune.	An R_0 value and a question about coverage.

Course scope (public description): Global and Population Health prepares students for the role of the nurse in preserving and promoting health among diverse populations. Additionally, basic principles of epidemiology, social determinants of health (SDOH), and resource allocation through value-based care are outlined. The course introduces planning, organization, and delivery of services for diverse populations in community settings, including illness prevention, disaster preparedness, and environmental health. All prior courses in the sequence for this program serve as prerequisites for this course. This course is eligible for an In Progress grade. Please see the Grading Scale Policy for more information.